

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-011-2012

Addendum

WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WS-004

WELDING PROCEDURE SPECIFICATIONS (WPS)

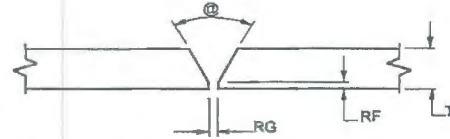
Company Name RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.
 Welding Procedure Specification no. RNC-WS-004
 Revision no. 0 Date -
 Welding Process(es) 1/ GTAW 2/ SMAW
 Impact Test Requirement ☐ Yes ☒ No

By Mr. Veerachat Fuenprayoon (Sign/Date)
 Date JULY 24, 2009
 Supporting PQR no.(s) RNC-PQ-004
 Type(s) 1/ Manual
 2/ Manual

JOINTS (QW-402)

Joint Design Butt joint (Single V-groove)
 Backing ☒ Yes for SMAW ☒ No for GTAW
 Backing Material (Type) 1/ NA
 2/ Weld metal (GTAW)
☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☒ Other (Weld Metal)
 Applicable of Tack Welding ☒ Yes ☐ No
 Applicable of Repair Welding ☒ Yes ☐ No
 Retainer None

Details



RG = 0 - 3.0 mm.
 RF = 0 - 2.0 mm.
 T = 5 - 36.52 mm.
 @ = 60° ± 5 deg.

BASE METALS (QW-403)

P-No. 5B Group no. All to P-no. 5B Group no. All
 OR Specification Type and Grade SA 335 Gr.P91 to SA 335 Gr.P91
 OR Chem. Analysis and Mech. Prop. NA to NA
 Base Metal Thk. Range (T, mm.) Groove 5.0 - 36.52 mm. Fillet All thicknesses
 Pipe Dia. Range Groove All diameters Fillet All thicknesses
 Other No t pass > 1/2 in.(13.0 mm.) for SMAW.

FILLER METALS (QW-404)

Spec. no. (SFA)	1/	5.28	2/	5.5	
AWS no. (Class)	1/	ER 90S-B9	2/	E9015-B9	
F-no.	1/	6	2/	4	
A-no.	1/	5	2/	5	
Size of Filler Metals	1/	Ø 2.4 mm.	2/	Ø 3.2 mm.	
Deposited Weld Metal	Thk. Range (t, mm.)				
	Groove	1/	16.0 mm (Max)	2/	20.52 mm (Max)
	Fillet	1/	Unlimited	2/	Unlimited
Type-wire	1/	Solid wire	2/	NA	
Electrode-Flux (class)	NA				
Flux Trade Name	NA				
Consumable Insert	None				
Other	Brand name : BOEHLER (ER 90S-B9), BOEHLER (E9015-B9)				

WPS no. RNC-WS-004 Rev. 0

POSITIONS (QW-405)

Position of Groove 1/ All 2/ All
Welding Progression : Up Hill Down -
Position(s) of Fillet 1/ All 2/ All

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range 730 - 760 °C
Time Range 1 hr/in. , Holding time 2.0 (Min) hr.
Other NA

PREHEAT (QW-406)

Preheat Temp. Min. 200 °C
Interpass Temp. Max. 350 °C
Preheat Maintenance Yes
Other See note 1)

GAS (QW-408)

	Percent Composition		
	Gas(es)	Mixture	Flow Rate
Shielding	<u>Argon</u>	<u>Welding Grade</u>	<u>10-20 L/min.</u>
Trailing	<u>NA</u>	<u>-</u>	<u>-</u>
Backing	<u>Argon</u>	<u>Welding Grade</u>	<u>10-25 L/min.</u>

ELECTRICAL CHARACTERISTICS (QW-409)

Current AC or DC 1/ DC 2/ DC Polarity 1/ EN 2/ EP
Amps (Range) See table below Volts (Range) See table below
Tungsten Electrode Size and Type Ø 2.4 mm / 2%Thoriated
Mode of Metal Transfer for GMAW/FCAW NA
Electrode Wire Feed Speed Range NA
Heat input (Range) NA

TECHNIQUE (QW-410)

String or Weave Bead 1/ String or Weave 2/ String or Weave
Orifice or Gas Cup Size 1/ No. 4 - 9 2/ NA
Initial and Interpass Cleaning Power brushing and / or Grinding
Method of Back Gouging NA
Oscillation NA
Contact Tube to Work Distance 1/ NA 2/ NA
Multiple or Single Pass (per side) 1/ Multiple 2/ Multiple
Multiple or Single Electrodes 1/ Single 2/ Single
Travel Speed (Range) See table below
Peening None
Other NA

Weld Layer(s)	Process(es)	Filler Metal		Current		Volt Range	Travel Speed Range: cm/minute	Heat Input Range : kJ/mm
		Class	Dia.	Type Polar.	Amp.Range			
1	GTAW	ER 90S-B9	2.4 mm.	DCEN	85-120 A	8-12 V	4 - 10	-
2	GTAW	ER 90S-B9	2.4 mm.	DCEN	100-165 A	10-14 V	4 - 10	-
3	GTAW	ER 90S-B9	2.4 mm.	DCEN	100-165 A	10-14 V	4 - 10	-
4 and Over	SMAW	E9015-B9	3.2 mm	DCEP	95-150 A	18-29 V	6 - 15	

Note : 1) Postheat shall be performed at 100 - 140 C° with holding time ½ hour (min) after complete welding immediately.



RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

59 M.8 HIGHWAY NO 36 T. TUBMA A MUANG RAYONG 21000

RENEWCOILS ENGINEER & SUPPLY CO.,LTD

Tel. (038) 663476 FAX. (038) 663477 Mobile (081) 8613795

Sheet 1 of 3

PROCEDURE QUALIFICATION RECORD (PQR)

Company Name RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Procedure Qualification Record no. RNC-PQ-004 Date JULY 24, 2009
WPS no. RNC-WS-004 Revision. no. 0
Welding Process(es) 1/ GTAW 2/ SMAW
Types (Manual, Automatic, Semi-Automatic) 1/ Manual 2/ Manual
Impact Test Requirement ☐ Yes ☒ No

JOINTS (QW-402)

SEE PAGE 3 OF 3

BASE METALS (QW-403)

Material Spec. SA335 to SA335
Type of Grade P91 to P91
P-no. 5B Group 2 to P-no. 5B Group 2
Thickness of Test Coupon, T 18.26 mm.
Diameter of Test Coupon NPS 10" SCH.100
Other No t pass > 1/2"(13.0 mm.) for SMAW.

POSTWELD HEAT TREATMENT (QW-407)

Temperature 760 °C
Time Holding Time 2 hr.
Other Heating Rate : 170 C° / hour
Cooling Rate : 185 C° / hour
(For more detail, see attached PWHT chart)

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	<u>Argon</u>	<u>99.90%</u>	<u>17 L/Min.</u>
Trailing	<u>NA</u>	<u>-</u>	<u>-</u>
Backing	<u>Argon</u>	<u>99.90%</u>	<u>20 L/Min.</u>

FILLER METALS (QW-404)

SFA Specification 1/ 5.28 2/ 5.5
AWS Classification 1/ ER 90S-B9 2/ E9015-B9
Filler Metal F.no. 1/ 6 2/ 4
Weld Metal Analysis A. No. 1/ 5 2/ 5
Size of Filler Metal 1/ Ø 2.4 mm. 2/ Ø 3.2 mm.
Other Solid wire filler metal for GTAW
Deposited Weld Metal 1/ 8.0 mm.
thickness, t (mm.) 2/ 10.26 mm.

ELECTRICAL CHARACTERISTICS (QW-409)

Current 1/ DC 2/ DC
Polarity 1/ EN 2/ EP
Amps. 1/ See sheet 3 2/ See sheet 3
Volts 1/ See sheet 3 2/ See sheet 3
Tungsten Electrode Size Ø 2.4 mm/2%Thoriated
Heat input (Range) NA
Other NA

POSITIONS (QW-405)

Position of Groove 1/ 6G 2/ 6G
Welding Progression (Uphill, Downhill)
1/ Uphill 2/ Uphill
Other None

TECHNIQUE (QW-410)

Travel Speed 1/ See sheet 3 2/ See sheet 3
String or Weave Bead 1/ Both 2/ Both
Oscillation NA
Multipass or Single Pass (per side) Multipass
Single or Multiple Electrodes Single
Other NA

PREHEAT (QW-406)

Preheat Temp. 200 °C
Interpass Temp. 350 °C
Other - Preheat temp. maintained by electric heater
- Postheat performed at temp 120 °C with
holding time 1 hour after complete welding
immediately.



RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

59 M.8 HIGHWAY NO. 36 T. TUBMA A. MUANG RAYONG 21000

RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Tel. : (038) 663476 FAX: (038) 663477 Mobile : (081) 8613795

Sheet 2 of 3

PQR Back

PQR no. RNC-PQ-004

TENSILE TEST (QW-150)

Specimen no.	Width (mm.)	Thickness (mm.)	Area (mm ²)	Ultimate Tensile Load (N)	Ultimate Unit Stress (N/mm ²)	Character of Failure and Location
TS 273 / 2009	19.76	18.54	366.35	254,500	694.69	Failure out of weld
TS 274 / 2009	19.90	18.46	367.35	256,500	698.24	Failure out of weld

GUIDED BEND TEST (QW-160)

Specimen no.	Type & Figure No. of Bend Tests	Results	Remarks
SB 1031 / 2009	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 1032 / 2009	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 1033 / 2009	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 1034 / 2009	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity

TOUGHNESS TEST (QW-170)

Specimen no.	Notch Location	Notch Type	Test Temp. °C	Impact Value			Drop Weight Break	
				J	%Shear	Mils	Break	No Break
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-

FILLET-WELD TEST (QW-180)

Results - Satisfactory : ☐ Yes ☐ No Penetration into Parent Metal : ☐ Yes ☐ No
Macro - Results : NA

OTHER TESTS

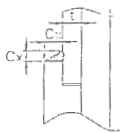
Type of Test RADIOGRAPHIC TEST AS PER REPORT NO.R09-3337-1-P42
Deposit Analysis NA
Other HARDNESS TEST (MIN/MAX/AVG.) : 206/254/230.9 HV AS PER REPORT NO.R09-3337-5-P42

Welder's name Mr. Suriyon Sanitnoi Clock no. - Stamp no. W - 005
Test conducted by SIWA TESTING INSPECTION & CONSULTING CO., LTD. Laboratory Test no. R09-3337-3,4-P42

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.

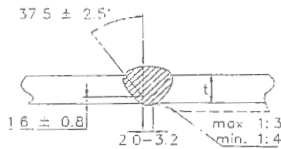
Manufacturer RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Date Certify _____

ATTACHED FIGURE JOINT SKETCH



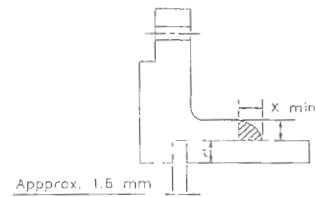
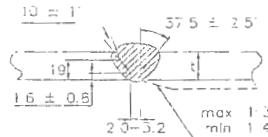
Cx (min.) = 1.25t but not less than 3.2 mm.

JOINT No. A1



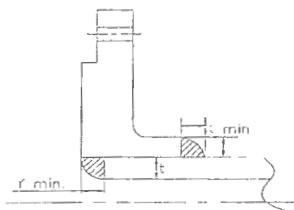
Misalignment (max.) = 1.5 mm

JOINT No. B1 & B2



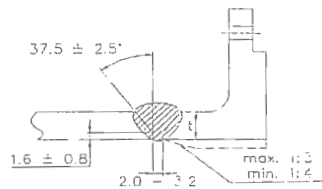
X min. = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



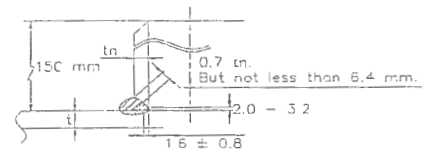
X min. = The lesser of 1.4t or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



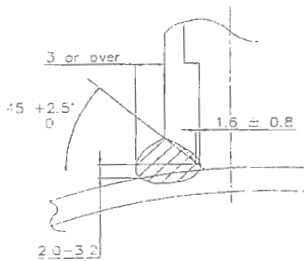
Misalignment (max.) = 1.5 mm.

JOINT No. B3

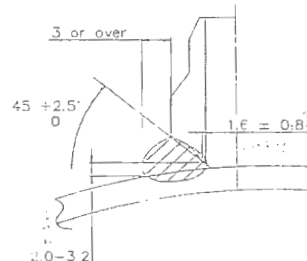


Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

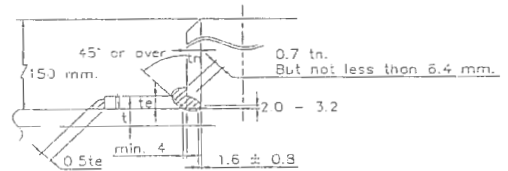
JOINT No. B4



JOINT No. C1

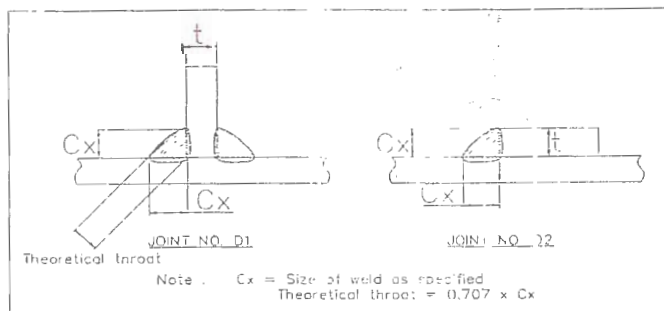


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm or 0.5tn

JOINT No. B5





บริษัท สิวา เทสติ้ง อินสเปกชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

ORIGINAL

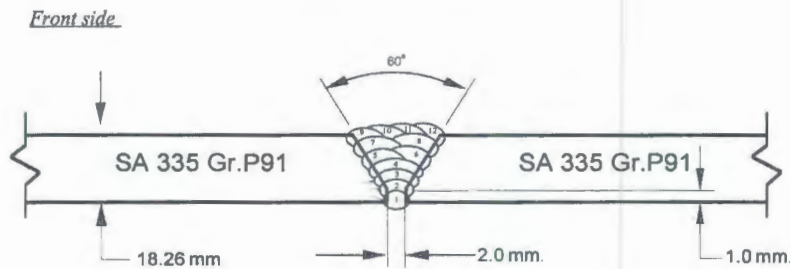
Sheet 3 of 3

Actual Records of Welding Variable

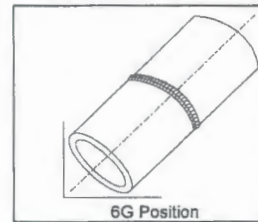
ACC. TO ASME SECTION IX WELDING VARIABLE

WPS NO. RNC-WS-004 Revision No. 0

WELDER NO. W-005



Type of material ☒ Pipe (NPS 10" SCH100) ☐ Plate
Backing ☒ Yes for SMAW ☒ No for GTAW
Retainers ☐ Yes ☒ No



PASS SEQUENCE

Side	Pass	Process	Electrode or Filler		Amps./ Polarity	Volts	Travel Speed (cm./minute)	Heat Input (kJ/mm)	Remarks (gas flow rate)
			Type	Size					
Front	1	GTAW	ER 90S-B9	Ø 2.4 mm.	100-105 / DCEN	10-10.5 V.	4.2	-	Ar: 17 l/Min.
	2	GTAW	ER 90S-B9	Ø 2.4 mm.	145-150 / DCEN	12-13 V.	8.3	-	Ar: 17 l/Min.
	3	GTAW	ER 90S-B9	Ø 2.4 mm.	145-150 / DCEN	12-13 V.	6.0	-	Ar: 17 l/Min.
	4	SMAW	E9015-B9	Ø 3.2 mm.	125-128 / DCEP	24-25 V.	10.2	-	-
	5	SMAW	E9015-B9	Ø 3.2 mm.	125-128 / DCEP	24-25 V.	9.7	-	-
	6	SMAW	E9015-B9	Ø 3.2 mm.	125-128 / DCEP	24-25 V.	9.9	-	-
	7	SMAW	E9015-B9	Ø 3.2 mm.	132-135 / DCEP	24-25 V.	11.3	-	-
	8	SMAW	E9015-B9	Ø 3.2 mm.	118-123 / DCEP	23-24 V.	10.5	-	-
	9	SMAW	E9015-B9	Ø 3.2 mm.	116-119 / DCEP	23-24 V.	10.7	-	-
	10	SMAW	E9015-B9	Ø 3.2 mm.	116-119 / DCEP	23-24 V.	9.4	-	-
	11	SMAW	E9015-B9	Ø 3.2 mm.	116-119 / DCEP	23-24 V.	11.8	-	-
	12	SMAW	E9015-B9	Ø 3.2 mm.	113-115 / DCEP	22-23 V.	13.2	-	-

Witnessed by:

J. Patiphan
(Mr. Patiphan)

Date: JUNE 30, 2009

FORM: PQR SECIX R.3 Page 3 of 3, 8 Jan. 2009

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.

BOOK NO. 425

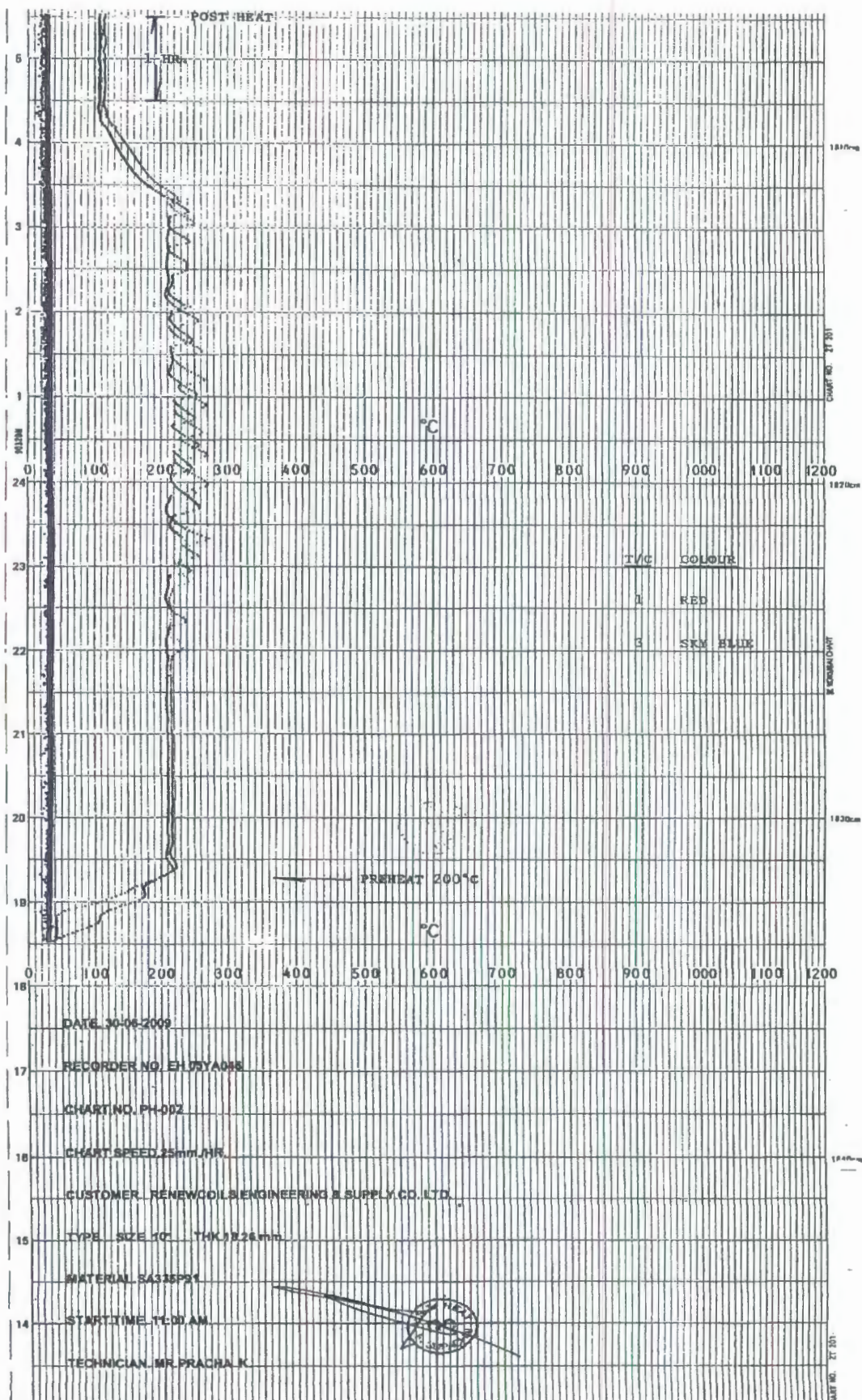
NUMBER 04

HEAT		J.S.T. HEAT TREATMENT COMPANY LIMITED	
PREHEATING REPORT			
CLIENT : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.			
CONTRACTOR : J.S.T.HEAT TREATMENT CO.,LTD.			
PROJECT :		PREHEAT METHOD: ☐ GAS ☒ ELECTRIC	
COMPLETION DATE : 30-06-2009		REPORT NO. PH-002	PAGE : 1
CHART SPEED : ☒ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.			
LOCATION : ☒ SHOP ☐ FIELD		RECORDER NO. EH 05YA046	
RESULT RECORDER CHART TO BE ATTACHED			
PREHEAT TEMPERATURE : 200		C./MIN	
INTERPASS TEMPERATURE : 350		C./MAX	
POSTHEAT TEMPERATURE : 100-121		C. SOAK 1 HR.	
MATERIAL : SA335P91			

LINE NO.& CLASS&SHT.NO&SPL.NO	WELD NO.	SIZE	THK (MM)	WELD TYPE	T/C NO.
PQR		10"	18.26	BW	1,3
TOTAL	1	JOINTS	10	D.B.	



CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.	CHECKED BY CUSTOMER	CHECKED BY CUSTOMER
SIGN : 	SIGN :	SIGN :
NAME : SUPACHAI K.	NAME :	NAME :
DATE : 30-06-2009	DATE :	DATE :



BOOK NO. 707

NUMBER 35

HEAT

J.S.T. HEAT TREATMENT COMPANY LIMITED

POST WELD HEAT TREATMENT REPORT

CLIENT: RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

CONTRACTOR : J.S.T. HEAT TREATMENT CO.,LTD.

PAGE : 1

PROJECT :

P.W.H.T.METHOD : ☒ LOCAL ☐ FURNACE

COMPLETION DATE: 02-07-2009

REPORT NO. PW-001

CHART SPEED : ☐ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.

RECORDER NO : EH 05YA046

MATERIAL : SA335P91

LOCATION : ☒ SHOP ☐ FIELD

UNIT :

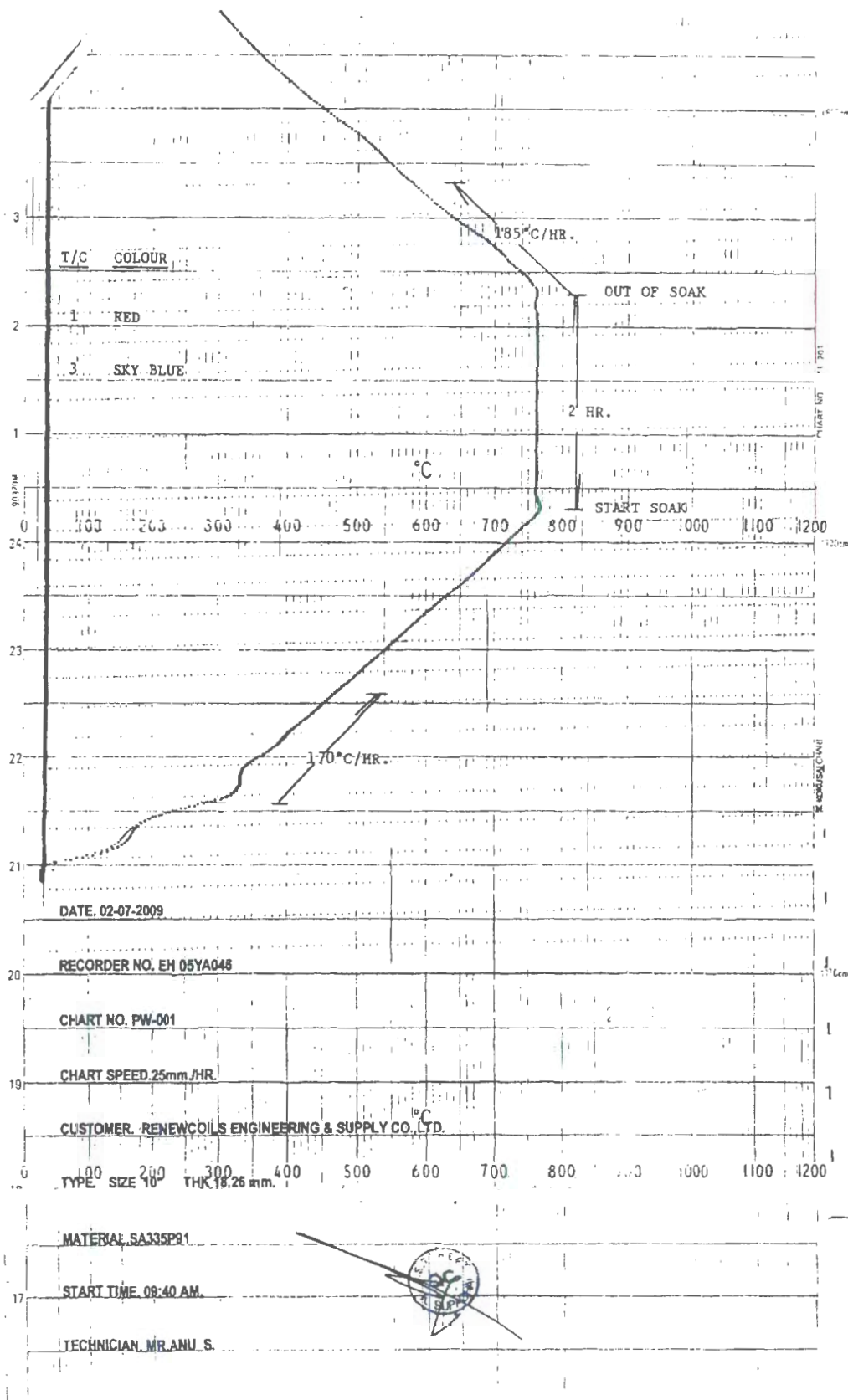
HEAT TREATMENT CONDITION/RESULTS

DESCRIPTION	LOADING TEMP(C°)	HEATING RATE (C°/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (C°/Hr)	UNLOADING TEMP (C°)
REQUEST	300	180	760 <u>±</u> 10	2 HR.	200	300
ACTUAL	300	170	760	2 HR.	185	300

[illegible]

TOTAL	1	JOINTS	TOTAL	10	D.B.
-------	---	--------	-------	----	------

CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.	REVIEWED BY	APPROVED BY	APPROVED BY
SIGN : 			
NAME : SUPACHAI K.			
DATE : 02-07-2009			



COOPERHEAT
Certificate of Calibration / Test



COOPERHEAT

INNOVATORS IN THERMAL TECHNOLOGY

Certificate No.

JST-073/2009

Date of Issue

8/2/2009

Signatory (Print)

SUPACHAI K.

Equipment Description

RECORDER "CHINO" EH SERIES

Input or Sensor Type

"K"

Output or Range

0-1200°C

Cooperheat Identification

Serial No.

Next Calibration Due

05YA046

8/8/2009

Customer

Customer Order No.

Works Order No.

Test Parameter	200	400	600	800	1000	1200
Prior to test	200	400	600	800	1000	1200
Post Test	200	400	600	800	1000	1200

Instrument Used	Model	Serial No.	Date Calibrated
FLUKE	714	6986010	20March2008

All measurements are performed with equipment that is traceable to National Standard unless otherwise stated

Signature

Date

Customer

8/2/2009

CERTIFICATE OF CALIBRATION



Issued By Marske Machine Calibration Centre

Date of Issue 20 March 2008

Cert Number

E80275

Page 1 of 3 Pages

Marske Machine (Thailand) Co., Ltd.
2nd Floor, The Seaboard Group Building
2/58 Highway 3191, Map Ta Phut, Muang, Rayong 21150
Tel : (038) 682992-3, (038) 691563-4
Fax : (038) 682988

Approved Signatory

Marut L. /TM

Customer : J.S.T. Heat Treatment Co., Ltd.
29/9 Soi Charoenmitr (Ekamai 10), Sukhumvit 63 Rd , Klongton Nue
Wattana, Bangkok 10110

Date Received : 19 March 2008

Instrument -	System ID	EC00745	Job Number	0000081246
	Description	Thermocouple Calibrator		
	Manufacturer	FLUKE		
	Model Number	714		
	Serial Number	6986010	Last Certificate Number	E70715
	Procedure Version	EP0008/7	Last Calibration Date	23-08-2007

Environmental Conditions

Temperature :	23°C +/- 2°C	Mains Voltage	220V +/- 10V
Relative Humidity :	55% +/- 15%	Mains Frequency	50Hz +/- 1Hz

Comments

The instrument was placed in lab and allowed to stabilise before calibrate.
Instrument set RJC external and use table IEC 584.1:1995/BS EN 60584 1:1996.
The direct comparison technique has been applied to calibrate.
All tests passed Mfg'Spec Model 714 Rev 1,10/97.

Traceability Information

Instrument description	Serial number	Certificate number	Cal. Date	Cal. Period
9822 Multifunction Calibrator	143411	12251	06-09-2007	52
3457A Multimeter	3114A15366	EL-0102/07	25-04-2007	52
M2001 Precision Calibration	M2001#2	063LTE-2	12-02-2008	52

Calibrated By : Suriya D.

Date of Calibration : 20 March 2008

This certificate provides traceability of measurement to recognised National Standards, and to the units of measurement realised at the National Physical Laboratory or other recognised National Standards laboratories.
Copyright of this certificate is owned by the issuing laboratory and may not be reproduced except with the prior written approval of the issuing laboratory.
This certificate complies with the requirements of B.S.5781, part 1 1992 (ISO 10012 - 1 1992).

Quality Control Executive Date of Issue

CERTIFICATE OF CALIBRATION



Cert Number
E80275

AS FOUND RESULTS

Page 2 of 3 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec	Uncertainties
------------	-----------	-----------	-------------	-----------	---------------

Calibration Information.

The uncertainty limits quoted refer to the measured values only, with no account being taken of the instrument ability to maintain its calibration.

The reported expanded uncertainty of measurement is state as the standard uncertainty of measurement multiplied by a coverage factor $k=2$ corresponding to a coverage probability of approximately 95%.

Function : DC Voltage Measurement

Range 75 mVDC	28.8uV	10.000mV	10.00mV	0%	7.6uV
Range 75 mVDC	28.8uV	-10.000mV	-10.00mV	0%	7.6uV
Range 75 mVDC	28.8uV	30.000mV	30.00mV	0%	7.7uV
Range 75 mVDC	28.8uV	50.000mV	50.00mV	0%	7.8uV
Range 75 mVDC	28.8uV	70.000mV	70.00mV	0%	8.1uV

Function : Temperature Measurement

1370 °C T/C Type K	0.83°C	-100.00°C	-100.1°C	12%	0.17°C
1370 °C T/C Type K	0.76°C	0.00°C	0.0°C	0%	0.16°C
1370 °C T/C Type K	0.74°C	100.00°C	100.0°C	0%	0.15°C
1370 °C T/C Type K	0.73°C	500.00°C	500.0°C	0%	0.15°C
1370 °C T/C Type K	0.76°C	1 000.00°C	999.9°C	13%	0.16°C
1200 °C T/C Type J	0.74°C	-100.00°C	-100.1°C	14%	0.15°C
1200 °C T/C Type J	0.70°C	0.00°C	-0.1°C	14%	0.14°C
1200 °C T/C Type J	0.68°C	100.00°C	99.9°C	15%	0.14°C
1200 °C T/C Type J	0.68°C	500.00°C	499.9°C	15%	0.14°C
1200 °C T/C Type J	0.67°C	1 000.00°C	999.9°C	15%	0.14°C
1760 °C T/C Type R	2.5°C	0.00°C	1.0°C	40%	0.50°C
1760 °C T/C Type R	1.8°C	100.00°C	100.0°C	0%	0.36°C
1760 °C T/C Type R	1.4°C	500.00°C	500.0°C	0%	0.28°C
1760 °C T/C Type R	1.3°C	900.00°C	900.0°C	0%	0.26°C
1760 °C T/C Type R	1.2°C	1 300.00°C	1 300.0°C	0%	0.24°C

Function : DC Voltage Source

Range 75 mVDC	28.8uV	9.990mV	10.00mV	35%	7.6uV
Range 75 mVDC	28.8uV	29.991mV	30.00mV	31%	7.7uV
Range 75 mVDC	28.8uV	49.989mV	50.00mV	38%	7.8uV
Range 75 mVDC	28.8uV	69.988mV	70.00mV	42%	8.1uV

Function : Temperature Source

1370 °C T/C Type K	0.83°C	-99.90°C	-100.0°C	12%	0.17°C
1370 °C T/C Type K	0.76°C	0.00°C	0.0°C	0%	0.16°C
1370 °C T/C Type K	0.74°C	99.90°C	100.0°C	14%	0.15°C
1370 °C T/C Type K	0.73°C	499.90°C	500.0°C	14%	0.15°C
1370 °C T/C Type K	0.76°C	999.90°C	1 000.0°C	13%	0.16°C
1200 °C T/C Type J	0.74°C	-99.90°C	-100.0°C	14%	0.15°C
1200 °C T/C Type J	0.70°C	0.00°C	0.0°C	0%	0.14°C
1200 °C T/C Type J	0.68°C	100.00°C	100.0°C	0%	0.14°C
1200 °C T/C Type J	0.68°C	500.00°C	500.0°C	0%	0.14°C
1200 °C T/C Type J	0.67°C	1 000.00°C	1 000.0°C	0%	0.14°C
1760 °C T/C Type R	2.5°C	-0.70°C	0.0°C	28%	0.50°C
1760 °C T/C Type R	1.8°C	99.10°C	100.0°C	50%	0.36°C
1760 °C T/C Type R	1.4°C	499.60°C	500.0°C	29%	0.28°C

Quality Isn't Expensive It's Priceless

CERTIFICATE OF CALIBRATION



Cert Number
E80275

AS FOUND RESULTS

Page 3 of 3 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainties
1760 °C T/C Type R	1.3°C	899.40°C	900.0°C	46%	0.26°C
1760 °C T/C Type R	1.2°C	1 299.80°C	1 300.0°C	17%	0.24°C

End of calibration.

Quality Isn't Expensive It's Priceless

STORK®
STORK®

Certificate of Conformance

This material has been tested, inspected and approved by Cooperheat quality control in compliance with Cooperheat's BS EN ISO 9001 : 2000 Quality Management System.

43000 (100 mtrs)

TYPE "K" (NiCr-NiAl) Twin Thermocouple wire conforms to:-

BS EN 60584-1:

BS EN 60584-2:

ANSI-MC96-1: Temperature Measurement Thermocouples. 1982

Batch No 1100804122

QC Inspector's Sig:



Date 03/10/2005





INSPECTION REPORT

CLIENT RENEW COILS ENGINEERING & SUPPLY CO., LTD.
PROJECT WPQ, WPS/PQR
INSPECTION DATE JUNE 29, 30 & JULY 1, 2009
REPORT NO. R09-3337-0-P42

SIWA TESTING INSPECTION & CONSULTING CO., LTD.

Bangkok Office

195 Soi Petchkasem 65, Petchkasem Road, Bangkhae, Bangkok 10160, Thailand

Tel. (66) (0) 2444-3645 (15 Lines) Fax. (66) (0) 2444-3026, 2444-3027 E-mail : info@siwatesting.com

Rayong Office

267/306-307 Sukhumvit Road, Map Ta Phut, Amphur Muang, Rayong 21150, Thailand

Tel. (66) (0) 3860-9620 (4 Lines) Fax. (66) (0) 3868-9394 E-mail : rayong@siwatesting.com

SUMMARY OF WELDER/WELDING OPERATOR PERFORMANCE QUALIFICATION

Manufacturer : RENEW COILS ENGINEERING & SUPPLY CO., LTD.
 Project : PQR
 Applicable Code : ASME SEC. IX Edition : 2007 Addenda : -
 Witness Date : JUNE 30, 2009 Place : RAYONG
 Material Specification : SA 335 Gr.P91
 Electrode/Filler Metal Classification : ER 90S-B9 + E 9015-B9

[illegible]

Legend : RT = Radiographic Test, NA = not applicable, ACC = Accept, REJ = Reject

Note. For complete qualification range, see attached certificate(s).

FORM : WQT-SUM2 R.1, 16 Mar. 2009

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.



RADIOGRAPHIC TEST REPORT

Report No. : R09-3337-1-P42

Page 1 of 1 ☐ Attached Report = Page(s)

Test Date : July 13, 2009

Client : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Source : ☐ X-ray ☒ Gamma Ray

Project :

Strength : 30 (kV/mA/Curies)

Product/Title : POR

Workplace : STIC

Film : ☐ Agfa ☐ Fuji ☒ Kodak

RT Procedure No. : PD-GEN-RT-ASME-01 REV.02

Type : AA

Technique : ☐ SWSI ☒ DWSI ☐ DWDI ☐ Panoramic

Penetrameter Placement :	UG : 0.020"
--------------------------	-------------

IQI Designation : ☐ DIN ☐ JIS ☒ ASTM 1B

☐ Source Side ☒ Film Side

Material : ☐ C/S ☐ A/S ☐ S/S ☒ ALLOY

Density : 2.0 - 4.0 Sensitivity : TABLE T-276

Welding Process : GTAW+SMAW

RT Operator : RCP

Total Films : $3.5'' \times 8.5'' =$ - $3.5'' \times 17'' =$ 4

Specification: ☐ ASME VIII ☒ ASME IX ☐ ASME B31.1 ☐ ASME B31.3

☐ AWS D1.1 ☐ API 1104 ☐[illegible]

Reviewed/Supervised by: PEO Date: 3/11/24 2009

Date : 5/11/24 2000

Client Approved :

Owner Approved :

(Udom Panpradab/RT Level II)

Certified by :

Rate :

Name :

Name :

(Somchai Ruangsuriyakij/Project Engineer)

24/7/2019

Date : _____

Date :

Signature :

Signature :

FORM - RT-STD2-POL R.1

ISSUE NO. 1

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.



บริษัท สิวา เทสติ้ง อินสเปกชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

Welder Cert. no. 336/2009

ORIGINAL

WELDER PERFORMANCE QUALIFICATIONS (WPQ)

Company's name : RENEW COILS ENGINEERING & SUPPLY CO., LTD.
Welder's name : Mr. Suriyon Sanitnoi Identification no. 3 5307 00363 19 0

Test Description

Identification of WPS followed or joint type RNC-WS-004 ☒ Test coupon ☐ Production Weld
Specification of Base Metal(s) SA 335 Gr. P91 Thickness 18.26 mm.
Filler metal or electrode specification (SFA) 5.28 + 5.5 Classification(s) ER 90S-B9 + E9015-B9

Testing Conditions and Qualification Limits

Welding Variables (QW-350)
Welding process(es) used
Type (ie; manual, semi-automatic) used
Backing (metal, weld metal, welded from both sides, flux, etc.)

Base metal (P-or S-Number) 5B Group 2 to (P-or S-Number)

☐ Plate ☒ Pipe (enter diameter, if pipe)

Groove
Fillet

Filler metal F-Number(s)

Filler metal product form (solid/cored/flux-cored) GTAW/PAW

Consumable insert for GTAW or PAW

Weld deposit thickness for each welding process

Process 1: GTAM 3 layers minimum ☒ Yes ☐ No

Process 2: SMAW 3 layers minimum ☒ Yes ☐ No

Welding position

Groove weld test

Plate & Pipe O.D. > 24" (610 mm)

Pipe O.D. ≤ 24" (610 mm)

Fillet weld test

Vertical Progression (uphill or downhill)

Backing gas for GTAW, PAW or GMAW/FCAW

GMAW/FCAW transfer mode

GTAW welding current type/polarity

Actual Values	Range Qualified
GTAW + SMAW	GTAW and/or SMAW
Manual	Manual
Without Backing for GTAW	With or without backing
With Backing for SMAW	With backing or back gouging
5B Group 2	P-or S-No. 1 through P-or S-No. 11 and metal of similar chem. composition to these metal
NPS 10" SCH 100	O.D. 2 7/8" to Unlimited
NA	All thicknesses & diameters
6+4	With or without backing : 6
Solid wire (GTAW)	With solid filler metal (GTAW)
None	With or without consumable insert
8.00 mm.	16.00 mm.(Max.) & all fillet sizes
10.26 mm.	20.52 mm.(Max.) & all fillet sizes
6G	All positions
NA	All positions
Uphill	Uphill
Argon 99.90 %	With backing gas
NA	-
DC/EN	DC/EN

RESULTS

Visual examination of Completed Weld (QW-302.4) ACCEPTABLE

Guided Bend Test (QW-160)

☐ Transverse root and face [QW-462.3(a)]; ☐ Longitudinal root and face [QW-462.3(b)]; ☐ Side [QW-462.2]
☐ Pipe bend specimen, corrosion-resistant [QW-462.5(c)] ☐ Plate bend specimen, corrosion-resistant [QW-462.5(d)]

Type and Specimen no.	Results
None	-
None	-

Type and Specimen no.	Results
None	-
None	-

Radiographic Examination (QW-191. For alternative qualification of groove welds by radiography)

Radiographic examination result (QW-191.2) ACCEPTABLE

Test conducted and evaluated by: SIWA TESTING INSPECTION & CONSULTING CO., LTD. Report no. R09-3337-1-P42

Fillet Weld Test (See QW-180)

Fracture test (QW-182) NA Length and percent of defects -
Macro Examination (QW-184) NA Fillet leg size (in.) - X - Concavity/convexity (in.) -

Other Tests

Type of test NA

Mechanical tests conducted and evaluated by: NA Report no. -

Welding Test Witnessed / Reviewed by: J. Patiphom, K. P. (Mr. Patiphan J. / Mr. Prasanna K.) Witnessed date: JUNE 30, 2009

SIWA TESTING INSPECTION & CONSULTING CO., LTD.

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of the Section IX of the ASME BOILER AND PRESSURE VESSEL CODE.

Certified by: (Manufacturer's representative) Date:

FORM WPQ-SEC0X R.5, 30 Apr. 2009

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.



TENSILE TEST REPORT

Report no. : R09-3337-3-P42 Page 1 of 1

Test Date : July 20, 2009

Client : Renew Coils Engineering & Supply Co., Ltd.

Tested Product : PQR No. RNC-PQ-004 (Welded Pipe $\varnothing$ 10")

Material : SA 335 Gr. P91

Project : -

Applicable Standard : ASME Sec. IX : 2007

Thickness : 18.26 mm

No.	Specimen No.	Size of Specimen (mm)	Gauge Length (mm)	Elongation (mm)	Elongation (%)	Tested Area (mm ²)	Load		Stress		Remark
							Yield (N)	Ultimate (N)	Yield (N/mm ²)	Ultimate (N/mm ²)	
1	TS 273 / 2009	19.76 x 18.54	-	-	-	366.35	-	254,500	-	694.69	Foot - Accept
2	TS 274 / 2009	19.90 x 18.46	-	-	-	367.35	-	256,500	-	698.24	Foot - Accept

Remark : FAW - Failure at weld
FOOW - Failure out of weld

Total	2	Specimen(s)	☐ Attached Report =	Page(s)
Operated by	Nopporn Karnjanajutaphan 		Reviewed and Supervised by :  (Patoomporn Phongchotip)	
Certified by	 (Patoomporn Phongchotip / Project Engineer)		Date : 23 JUL 2009	

FORM Tensile Test R.13Jan07

ISSUE NO. 1

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม 65 ถนนเพชรเกษม แขวง/เขตบางกะปิ กรุงเทพฯ 10160
Bangkok Office 195 Soi Petchkasem 65, Petchkasem Road, Bangkokae, Bangkok 10160, Thailand
Tel. (66)(0) 2444-3645 (15 lines) Fax. (66)(0) 2444-3028, 2444-3027 Email. info@siwatesting.com

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท ต.นาบตาพุด อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Road, Map Ta Phut, Amphur Muang, Rayong 21150, Thailand
 Tel. (66)(0) 3860-9620 (4 Lines) Fax. (66)(0) 3868-9394 E-mail. rayong@siwatesting.com Website. www.siwatesting.com

ORIGINAL 1



GUIDED BEND TEST REPORT

Report no. : R09-3337-4-P42 Page 1 of 1

Test Date : July 20, 2009

Client : Renew Coils Engineering & Supply Co., Ltd.

Project : -

Tested Product : PQR No. RNC-PQ-004 (Welded Pipe Ø 10")

Material Specification : SA 335 Gr. P91

Thickness :	18.26	mm
-------------	-------	----

Applicable Standard : ASME Sec. IX : 2007

Bending Angle : 180

Deg.	Mandrel Dia. :	38.0	mm
------	----------------	------	----

Specimen No.	Type of Bending	Size of Specimen	Fracture Location	Remark
SB 1031 / 2009	Transverse Side Bend	9.5 x 18.26	No discontinuity	Accept
SB 1032 / 2009	Transverse Side Bend	9.5 x 18.26	No discontinuity	Accept
SB 1033 / 2009	Transverse Side Bend	9.5 x 18.26	No discontinuity	Accept
SB 1034 / 2009	Transverse Side Bend	9.5 x 18.26	No discontinuity	Accept

[illegible]

Total :	4	Specimen(s)	☐ Attached Report =	Page(s)
---------	---	-------------	--	---------

Operated by	Nopporn Karnjanajutaphan / 
-------------	--

Reviewed and Supervised by : Patoorn
(Patoomporn Phongchotip)

Certified by	  (Patoomporn Phongchotip, Professional Engineer)
--------------	---

Date : 23 JUL 2009

FORM : Bend Test R.13Jul07

ISSUE NO. 1

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม 65 ถนนเพชรเกษม แขวง/เขตบางเขน กรุงเทพฯ 10180
Bangkok Office 195 Soi Petchkasem 65, Petchkasem Road, Bangkok
Bangkok 10180, Thailand
Tel. (66)(0) 2444-3645 (15 lines) Fax: (66)(0) 2444-3026, 2444-3027
Email: info@slwatesting.com Website: www.slwatesting.com

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Road, Map Ta Phut,
Amphur Muang, Rayong 21150, Thailand
Tel. (66/0) 3660-9620 (4 Lines) Fax. (66/0) 3668-9394
E-mail. rayong@slwatesting.com Website. www.slwatesting.com

ORIGINAL.



บริษัท ศิวะ เทสติ้ง อินสเปคชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

HARDNESS TEST REPORT

Report no. : R09-3337-5-P42 Page 1 of 1

Test Date : July 20, 2009

Client : Renew Coils Engineering & Supply Co., Ltd.

Project : -

Tested Product : PQR No. RNC-PQ-004 (Welded Pipe Ø 10")

Stage of Test ☐ Material Verification ☒ As Welded
☐ After PWHT ☐ Other

Test Product :

☐ Plate ☒ Pipe
☐ Weld ☐ Others

Material Specification SA 335 Gr. P91

Thickness 18.26 mm

Equipment WOLPERT HARDNESS TESTER

Indenter Size STANDARD PYRAMID DIAMOND (136°)

Test Load 10.0 kg.

Applicable Standard ASME Sec. IX : 2007 / E92

Test Location



H. 62 / 2009

Point No.	Location	Average Diameter/ Diagonal	Hardness Value		Point No.	Location	Average Diameter/ Diagonal	Hardness Value	
			Brinell (HBN)	Vickers (HV)				Brinell (HBN)	Vickers (HV)
1	BASE	0.290	-	221.0	22	WELD	0.270	-	254.0
2	BASE	0.290	-	221.0	23	WELD	0.270	-	254.0
3	BASE	0.290	-	221.0	24	WELD	0.270	-	254.0
4	HAZ	0.290	-	221.0	25	HAZ	0.270	-	254.0
5	HAZ	0.280	-	236.0	26	HAZ	0.270	-	254.0
6	HAZ	0.300	-	206.0	27	HAZ	0.280	-	236.0
7	WELD	0.280	-	236.0	28	BASE	0.280	-	236.0
8	WELD	0.280	-	236.0	29	BASE	0.290	-	221.0
9	WELD	0.280	-	236.0	30	BASE	0.290	-	221.0
10	HAZ	0.290	-	221.0	*****				
11	HAZ	0.290	-	221.0					
12	HAZ	0.300	-	206.0					
13	BASE	0.290	-	221.0					
14	BASE	0.290	-	221.0					
15	BASE	0.290	-	221.0					
16	BASE	0.300	-	206.0					
17	BASE	0.290	-	221.0					
18	BASE	0.285	-	228.0					
19	HAZ	0.270	-	254.0					
20	HAZ	0.280	-	236.0					
21	HAZ	0.270	-	254.0					

Total : 30

Point(s)

☐ Attached Report = Page(s)

Operated by : Nopporn Karnjanajutaphan

Reviewed and Supervised by :

(Patoomporn Phongchotip)

Certified by :

(Patoomporn Phongchotip) Project Engineer

Date :

23 JUL 2009

FORM : Hardness R.13Jul07

ISSUE NO. 1

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม 65 ถนนเพชรเกษม แขวง/เขตบางพลัด กรุงเทพฯ 10160
Bangkok Office 195 Soi Petchkasem 65, Petchkasem Road, Bangkok
Bangkok 10160, Thailand
Tel. (66)(0) 2444-3845 (15 lines) Fax. (66)(0) 2444-3026, 2444-3027
Email. info@siwatesting.com Website. www.siwatesting.com

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Road, Map Ta Phut,
Amphur Muang, Rayong 21150, Thailand
Tel. (66)(0) 3860-9820 (4 Lines) Fax. (66)(0) 3868-8394
E-mail. rayong@siwatesting.com Website. www.siwatesting.com